



KEVIN EKA PRATAMA

BACKEND & CLOUD SYSTEMS ENGINEER

Distributed Microservices • High-Concurrency Go & Java Spring Boot • ACID Ledgers

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EXECUTIVE SUMMARY

TARGET: BACKEND (PAGE 1 OF 3)

Backend & Cloud Systems Engineer with 2+ years of enterprise Application Support experience at PT PLN Icon+. Proven track record maintaining 100% SLA compliance for production operational tickets, authoring complex SQL queries (PostgreSQL, Oracle, MySQL) for data auditing, and monitoring 24/7 high-availability system workflows. Deeply proficient in AI-Assisted Engineering, pairing with LLM tools to accelerate schema design, boilerplate generation, and integration testing. Independently architected and deployed 19 production-grade backend microservices and cloud systems using Go (Golang), Java Spring Boot 3.3, Bun/Hono, and Express.js. Strong mastery in Clean Architecture (DDD), ACID transactional ledgers with row-level locks, Redis cache-aside patterns, RabbitMQ message brokers, gRPC binary protocols, and Docker containerization.

PROFESSIONAL EXPERIENCE

Application Support Engineer • PT PLN Icon+

2023 - Present

- Investigated and resolved technical operational tickets with a **100% SLA compliance rate**, ensuring timely resolution of customer transaction issues.
- Authored and executed complex SQL queries across **PostgreSQL, Oracle, and MySQL** for operational data validation, transaction auditing, and business reporting.
- Monitored nationwide enterprise system workflows 24/7, analyzed application error logs (HTTP 5xx/4xx), and coordinated directly with core developers for bug/API fixes.

AI-Assisted Backend Systems & Architecture • Independent Engineering & Open Source Projects

2023 - Present

- Pioneered AI-assisted software engineering workflows (Gemini 2.5, Claude 3.7, OpenAI, agentic coding tools) for rapid architectural scaffolding, schema design, and automated test suite generation.
- Architected, built, and audited **19 production-grade backend microservices** with Clean Architecture and Docker containerization.
- Audited AI-generated architectures for strict security, PostgreSQL row-level locks, Redis cache-aside patterns, and RabbitMQ decoupled message brokers.

TECHNICAL COMPETENCIES & CORE STACK

Backend Languages: Go (Golang 1.25/1.26), Java (JDK 17/21), TypeScript, JavaScript (Node.js/Bun), Python 3, SQL, Bash

Frameworks & Architecture: Java Spring Boot 3.3 (Spring Security 6, JPA), Go (Gin/Fiber/Echo), Bun + Hono, Express.js, FastAPI, Clean Architecture (DDD), gRPC (Protobuf), RESTful APIs, Microservices

Databases & Message Brokers: PostgreSQL (GORM, Prisma, ACID Transactions, Connection Pooling, Row-level Locks), MySQL, MongoDB (NoSQL), Redis (Cache-Aside, Rate Limiting), RabbitMQ (Message Broker)

AI & Agentic Engineering: Gemini 2.5, Claude 3.7, OpenAI APIs, AI Component Prototyping, Prompt Engineering, Agentic Tooling, Automated Testing

DevOps, Cloud & Tooling: Docker, Docker Compose, Linux Bash, Git & GitHub, Postman, Swagger / OpenAPI 3.0, CI/CD GitHub Actions

EDUCATION

Bachelor of Computer Science / Information Technology (S.Kom) • Universitas AMIKOM • GPA: 3.42 / 4.00 (Thesis Defense: Dec 2022 | Official Degree / Graduation: 2023)

2017 -
2023

19 Repos

BACKEND REPOSITORIES

8 Microservices

GO (GOLANG) SERVICES

4 Services

JAVA SPRING BOOT

7 Services

NODE.JS / BUN / PYTHON

PILLAR 1: DISTRIBUTED GO & JAVA HIGH-THROUGHPUT ENGINES

6 Repositories

- go-distributed-microservices-lab: High-throughput microservices with binary gRPC, Protocol Buffers, RabbitMQ event bus, and Redis cache-aside.
- go-ecommerce-gateway-engine: High-performance API Gateway with Gin router, MongoDB v2, order lifecycle, reverse proxy, and Swagger docs.
- go-banking-core-system: Core banking ledger with atomic balance transfers, ACID PostgreSQL row locks, Bcrypt PIN, and Swagger docs.
- go-clean-arch: Domain-Driven Design (DDD) Clean Architecture decoupling Domain, Usecase, and Repository data layers.
- go-rest-api-enterprise: Enterprise Go REST API with Gin, GORM, Redis caching, Uber Zap structured logging, and graceful shutdown.
- spring-boot-enterprise-platform: Java 17 Spring Boot 3.3 enterprise platform with Spring Security JWT, AOP logging, Bucket4j rate limiting, and Docker.

PILLAR 2: CLOUD APIS & TYPESCRIPT MICRO-FRAMEWORKS

6 Repositories

- hono-ecommerce-engine: Sub-millisecond REST API engine running on Bun runtime with Hono v4, Drizzle ORM, and WebSocket live chat.
- express-prisma-realworld-api: RealWorld standard backend with Express, TypeScript, Prisma ORM, Nx Monorepo, JWT, and Jest test suite.
- express-typescript-prisma-api: Type-safe REST API built with Express v5, TypeScript, Prisma 7 ORM, LibSQL adapter, and tsx development.
- express-prisma-product-api: REST API with Express v5, Prisma ORM, JWT authentication, Multer upload, and Zod runtime validation.
- express-sqlite-ecommerce-api: Lightweight e-commerce API with Express v5, SQLite prepared statements, ACID checkout, and Swagger UI.
- express-realtime-api-service: Express v5 with Socket.IO real-time broadcasting, dual database (Mongoose + MySQL), Winston logger, and Zod.

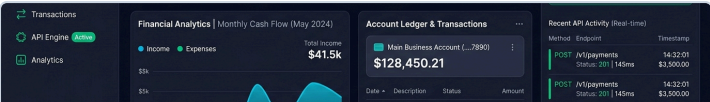
PILLAR 3: SPECIALIZED MICROSERVICES, WEBHOOKS & CACHING

7 Repositories

- express-prisma-payment-api: Payment backend with Midtrans webhook verification, automated PDFKit digital invoice rendering, and email.
- AI-api-manager: Reverse proxy API gateway with API key authentication, distributed rate limiting, token quota tracking, and React UI.
- spring-boot-book-manager-api: Java 17 and Spring Boot 3.3 REST service with Spring Data MongoDB, pagination, and OpenAPI docs.
- express-book-catalog-api: Book catalog API with Express v5, Prisma 7, Socket.IO live notifications, Redis rate limiting, and Jest tests.
- express-redis-url-shortener: URL shortener engine with Redis cache-aside pattern for sub-millisecond redirects and MongoDB persistence.
- express-mongo-content-api: Content management REST API with Express, MongoDB, Redis caching, Socket.IO broadcasting, and cron jobs.
- express-mongodb-starter-api: Modular CRUD boilerplate with Express, MongoDB Mongoose, JWT auth, Redis caching, and Docker container.

Directory Audit Note: Full source code, commit history, and test suites for all repositories are publicly available at github.com/mazkev and interactive web workstation at mazkev.vercel.app.

Interactive API Contracts, Microservices Topologies, Schema Proofs & Rate Limiters



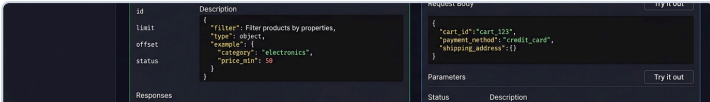
1. GoFinance Banking Core API

BACKEND

Go • Echo • PostgreSQL • Redis • RabbitMQ • Docker

High-concurrency banking engine with ACID transactional transfers, Redis cache-aside ledger, and RabbitMQ broker.

github.com/mazkev/go-banking-core-system



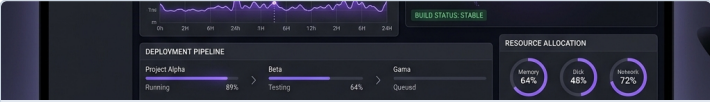
2. Swagger Go API Gateway Engine

BACKEND

Go 1.26 • Gin • GORM • PostgreSQL • Swagger OpenAPI

Production API gateway with interactive Swagger OpenAPI contract documentation, connection pooling, and JWT authorization.

github.com/mazkev/go-ecommerce-gateway-engine



3. Nexus Enterprise Microservices

BACKEND

Java Spring Boot 3.3 • Resilience4j • Eureka • PostgreSQL

Enterprise backend architecture with Eureka service discovery, circuit-breaker failover protection, and JPA auditing.

github.com/mazkev/nexus-workspace-engine



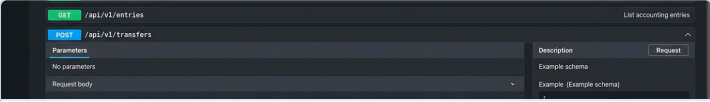
4. Go Clean Architecture Engine

BACKEND

Go • Gin • Clean Architecture • Docker • Unit Tests

Modular domain-driven design decoupling business logic, usecases, and repository data stores for maximum testability.

github.com/mazkev/go-clean-arch



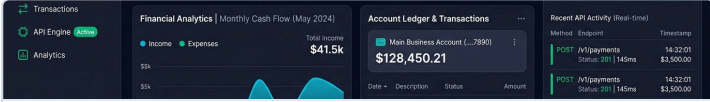
5. Core Banking Swagger UI & Ledger

BACKEND

Go • Echo • Swagger UI • Bcrypt PIN • Audit Logs

Interactive API testing suite verifying balance inquiries, atomic debit/credit transactions, and audit ledger entries.

github.com/mazkev/go-banking-core-system



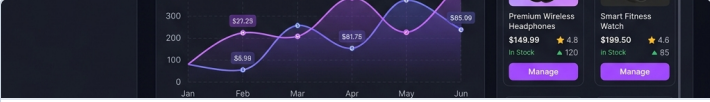
6. Microservices Concurrency Lab

BACKEND

Go • gRPC • Protocol Buffers • RabbitMQ • Redis

Binary inter-service communication pipeline with worker pool concurrency and decoupled background message queues.

github.com/mazkev/go-distributed-microservices-lab



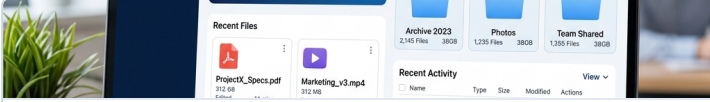
7. Bun Hono Ultra-Fast REST Engine

BACKEND

Bun • Hono v4 • Drizzle ORM • TypeScript • WebSocket

Sub-millisecond REST API engine running on Bun runtime with Drizzle ORM, live WebSocket chat, and coupon discount logic.

github.com/mazkev/hono-ecommerce-engine



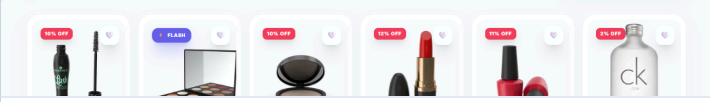
8. AI API Manager & Rate Limiter

BACKEND

Node.js • Express • Redis • Token Quotas • React Console

Reverse proxy API gateway with API key authentication, distributed rate limiting, token quota tracking, and latency analytics.

github.com/mazkev/AI-api-manager



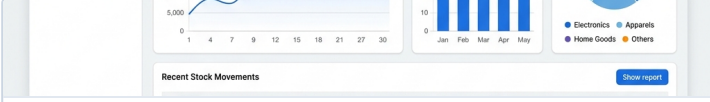
9. Midtrans Payment Gateway API

BACKEND

Node.js • Express v5 • Prisma ORM • Midtrans • PDFKit

Payment processing backend with Midtrans webhook verification, automated digital PDF invoice rendering, and email notifications.

github.com/mazkev/express-prisma-payment-api



10. Spring Boot Enterprise Platform

BACKEND

Java 17 • Spring Boot 3.3 • JWT • Bucket4j • MongoDB

Enterprise platform with Spring Security JWT, AOP audit logging, async event-driven mailers, Bucket4j rate limiting, and Docker.

github.com/mazkev/spring-boot-enterprise-platform

Interactive Demonstration & Source Code Audit: Live deployments, interactive case studies, architectural documentation, and full source code are accessible at mazkev.vercel.app and github.com/mazkev.